CCU046: Severe mental illness and receipt of acute cardiac care and mortality following myocardial infarction

Output 2: Mental illness and mortality following myocardial infarction

Protocol

Version 1.0

Version control

Version	Date	Author(s)	Comment
0.1	3 September 2024	Kelly Fleetwood Caroline Jackson	Initial draft
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Summary

Severe mental illness (SMI), which includes schizophrenia, bipolar disorder and major depression, affects roughly one in ten adults every year in the United Kingdom (1). People with SMI die 10-20 years earlier than people without SMI (2, 3). This is mainly due to poorer physical health, in particular a higher risk of cardiovascular disease (CVD), including myocardial infarction (MI) (2, 4).

After an MI, people with SMI are more likely to die than those without SMI (5). The reasons for this difference in outcome remain unclear and require deeper investigation, including to determine the role of confounding variables such as key comorbidities, as well as the role of receipt of care. The contribution of the latter to the difference in mortality outcome has not yet been explored in depth. Furthermore, we do not know whether any differences in risk of dying after an MI have been affected by the COVID-19 pandemic. It is important to investigate this, to inform responses to this now and in future situations where disruptions of care occur.

This protocol describes the methods we will use to examine SMI and mortality following MI. Separate protocols address SMI and receipt of acute cardiac care following MI (CCU046_01) and incidence of MI, stroke and heart failure by SMI (CCU046_03).

Objectives

(1) Among people with an MI:
   a) is comorbid SMI associated with excess 30-day and one-year mortality?
   b) how has the COVID-19 pandemic affected SMI differences in 30-day and one-year mortality?

Data

Data for this project will come from the CVD-COVID-UK/COVID-IMPACT consortium (17). The data includes individual health records from England from multiple healthcare settings, including GP data (for people alive on 1 November 2019), hospital data, CVD clinical audit data and COVID-19 test results.

Specifically, we will use the following datasets from NHS Digital's Trusted Research Environment (TRE) for England for the CVD-COVID-UK/COVID-IMPACT consortium:

1. GDPPR: GPES Data for Pandemic Planning and Research
2. HES: Hospital Episode Statistics
3. COVID-19 SGSS: Second Generation Surveillance System
4. Civil Registration – Deaths
5. Myocardial Ischaemia National Audit Project (MINAP)

A description of each of these datasets and the data that we will require is included below.

Myocardial Ischaemia National Audit Project (MINAP)

MINAP is an audit of patients admitted to hospital in England, Wales and Northern Ireland with MI (data from England is available in the TRE for England). It is administered by NICOR. MINAP seeks to include all hospitalised STEMI or NSTEMI events, with participating centres submitting information from across the patient journey, from the point of call to ambulance service or self-presentation at hospital, through diagnosis and treatment, to medication prescribed at discharge (18). Case ascertainment in MINAP is considered to be very good, with the most recent median case ascertainment in England reported as 99%. The MINAP collects a substantial breadth and depth of information on 130 data items, including patient characteristics and medical history, smoking status,

various physiological and biological measures, diagnostic tests, investigations and interventions received (including reasons for non-receipt), and medications on admission and discharge. Information on time intervals/delays are also collected (i.e. time from patients' call for help to receipt of PCI [call-to-balloon] and arrival at hospital [call-to-door] and from arrival at hospital to receipt of PCI [door-to-balloon]).

The CVD-COVID-UK/COVID-IMPACT resource includes MINAP data for admissions from January 2017. Currently, records appear to be complete up to March 2023. We will use MINAP to define the cohorts and identify key factors associated with MI survival, including MI presentation and receipt of care, that might differ by SMI status.

GDPPR: General Practice Extraction Service (GPES) Data for Pandemic Planning and Research

The GDPPR dataset (https://digital.nhs.uk/coronavirus/gpes-data-for-pandemic-planning-and-research/guide-for-analysts-and-users-of-the-data) includes GP records from individuals with active, current registrations at participating practices and deceased patients with a date of death on or after 1 November 2019. Individuals who opted out of secondary use of their GP data via the National Data Opt-out scheme are excluded from the dataset.

Clinical records are coded using SNOMED-CT codes. The GDPPR dataset does not include all codes recorded in primary care. It includes a selection related to specific diagnoses, prescriptions and lifestyle factors. Some codes are available from the start of each individual's records, however a subset are only available for clinical records within the last two years. Codes for MI, schizophrenia, bipolar disorder and depression are available from the start of each individual's records.

We will use primary care data to ascertain history of severe mental illness in some analyses. Currently, records appear to be complete up to October 2023.

HES APC: Hospital Episode Statistics Admitted Patient Care

The HES APC dataset includes records of all regular NHS hospital treatment in England. It excludes records from accident and emergency departments and from outpatient appointments. The dataset includes information about the date of admission, the primary diagnosis, up to 19 secondary diagnoses and operations, amongst other variables. Diagnoses are coded using the International Classification of Diseases 10th revision (ICD-10).

We will use HES data to identify severe mental illness. The CVD-COVID-UK/COVID-IMPACT resource includes HES data for all completed episodes of care from 1997. Currently, records appear to be complete up to October 2023.

COVID-19 SGSS: Second Generation Surveillance System

The COVID-19 SGSS dataset includes data on individuals who tested positive for COVID-19. We will use this data to identify COVID-19 status immediately prior to and after MI.

The CVD-COVID-UK/COVID-IMPACT resource includes SGSS data for admissions from the start of records in 2020. Currently, records are available up to October 2023.

Civil Registration – Deaths

The death records include all deaths registered in England or Wales. The data includes information on the date of death, the primary cause of death and up to 15 secondary causes of death. Causes of death are coded using ICD-10 codes.

We will use death records to identify deaths following myocardial infarction.

The CVD-COVID-UK/COVID-IMPACT resource includes death records from January 1993. Currently, records appear to be complete up to October 2023.

Methods

Cohorts

MI can be classified as ST segment elevation (STEMI) or non-ST segment elevation (NSTEMI). A STEMI is caused by a complete blockage of the coronary artery and is more severe than an NSTEMI which is caused by a partial blockage of the coronary artery. We will create separate cohorts for patients with STEMI and NSTEMI.

For each type of MI, we will conduct mortality analyses using two cohorts – the primary cohort will include MIs from 1 November 2019, and will use both primary care and hospital data to define SMI. The secondary cohort will include MIs from 1 January 2017, but will only use hospital data to define SMI (since primary care data is only available for people alive on 1 November 2019).

Primary cohort: Adults with a STEMI or NSTEMI recorded in MINAP from 1 November 2019 to latest date of complete MINAP records

Our primary cohort will include adults (≥ 18 years of age) in the GDPPR dataset with a final diagnosis of STEMI or NSTEMI recorded in MINAP from 1 November 2019 to latest date of complete MINAP records. For each individual we will include the first record of an MI within the study period.

Individuals in this dataset were, by definition, alive on the 1 November 2019. Thus we expect to have GP records for the majority of these individuals from their earliest GP record up to their most recent GP record. For these individuals we will be able to identify history of SMI based on a combination of both their GP records and their hospital records.

Secondary cohort: Adults with a STEMI or NSTEMI recorded in MINAP from 1 January 2017 to latest date of complete MINAP records

Our secondary cohort will include adults (≥ 18 years of age) with a final diagnosis of STEMI or NSTEMI recorded in MINAP from 1 January 2017 to latest date of complete MINAP records. For each individual we will include the first record of an MI within the study period.

This cohort provides us with a longer period of pre-pandemic MI data. It will give us more data to evaluate whether the COVID-19 pandemic affected any SMI differences in mortality. However, GP records are not available for people who died prior to 1 November 2019, hence for this cohort

history of SMI will be based only on hospital data. We will therefore under-ascertain the proportion of people in the dataset with a history of SMI.

Exposure: severe mental illness

We will ascertain schizophrenia, bipolar disorder and depression from primary care and hospital data. Whilst the primary care data has an advantage over hospital data in terms of allowing ascertainment of a history of a clinical diagnosis of these mental illnesses, these data do not differentiate severity of depression. This means that we will identify people with a diagnosis of depression across the severity spectrum and not just major depression. We will however investigate the impact of this ascertainment method on results in our secondary analyses, where we define SMI from hospital admission records only.

For the primary care data, SMI will be identified using systematized Nomenclature of Medicine Clinical Terms (SNOMED CT) codes.

For the hospital data, SMI will be identified using ICD-10 codes. We will identify SMI based on both primary and secondary diagnoses.

For analysis, we will define mutually exclusive groups of people with schizophrenia, bipolar disorder or depression. We will categorise people with a history of more than one SMI according to their most severe illness, with schizophrenia considered the most severe, followed by bipolar disorder and depression. Our analyses will compare people in each of these three groups to a control group. The precise definition of our control group will be discussed in our first collaborator meeting in December 2022, but will likely be either a) people without a prior diagnosis of SMI, or b) people without a prior diagnosis of any mental illness.

Outcomes

The primary outcome is 30-day mortality.

Secondary outcomes include:

- one-year all-cause mortality (for people where death data is available for at least one year post-MI date)
- one-year cardiovascular mortality (for people where death data is available for at least one year post-MI date) and
- time to all-cause mortality and to cardiovascular death in the first year post MI (we are interested in mortality at 1 year and not beyond given that some of our covariates such as the receipt of care may influence mortality at 1-year but are less likely to influence mortality at 5-years)

We will use ICD-10 codes to identify cardiovascular mortality from the primary cause of death (see codes below).

Cause-specific mortality	ICD-10 codes	Description
CVD-specific mortality	I20-I25	Ischaemic heart diseases
	I60-I69	Cerebrovascular diseases
	G45	Transient cerebral ischaemic attacks and related syndromes

Covariates

In the analyses of mortality, we will serially adjust models for confounding factors to assess the effect of adjustment for groups of factors on effect estimates. These include hospital, month and

year of admission, timing of admission (weekday or weekend, daytime or overnight), sociodemographic factors (age, sex, ethnicity and area-based deprivation), lifestyle factors (smoking and body mass index), existing CVD (previous MI, angina), previous PCI, comorbidities (chronic cardiac failure, diabetes, chronic renal failure, cerebrovascular disease), MI presentation (such as cardiogenic shock, ST-segment changes, cardiac arrest, troponin elevation, COVID-19 status), and receipt of care standards.

Statistical analysis

We will analyse patients with STEMI and NSTEMI separately.

Data quality assessment

We will review the completeness of all variables that we plan to use in analyses and also cross-check variables against each other where possible. We will also examine how COVID-19 testing changed over time, as community testing was introduced, which will inform how we will account for COVID-19 infection in the analyses.

Descriptive analyses

We will compare patient and event characteristics among people with versus without SMI. We will examine whether characteristics of patients, and mortality changed during the COVID-19 pandemic. We will also assess variable missingness

Objective 1: Do people with an SMI experience any differences in acute cardiac care?

In order to address objective 1, we will use our primary cohort, as defined above.

We will analyse mortality outcomes in those with versus without SMI using logistic regression analyses or Cox proportional hazards regression depending on whether we are analysing occurrence of event or time to event. We will use a competing risks model where appropriate (e.g. accounting for competing risk of deaths from other causes when analysing time to CVD mortality). Models will be adjusted for covariates including patient and MI event characteristics. We will use multiple imputation by chained equations to handle missing data in covariates, where appropriate.

Objective 2: How have any differences in acute cardiac care been affected by the COVID-19 pandemic?

In order to address objective 2, we will use our secondary cohort, as defined above. To examine the effect of the COVID-19 pandemic on associations between SMI and outcomes, we will include continuous calendar time in the models and evaluate interactions between continuous calendar time and SMI. This will allow us to determine the effect of the pandemic on MI outcomes and care without imposing a somewhat artificial cut-point for “pre” versus “post” pandemic.

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